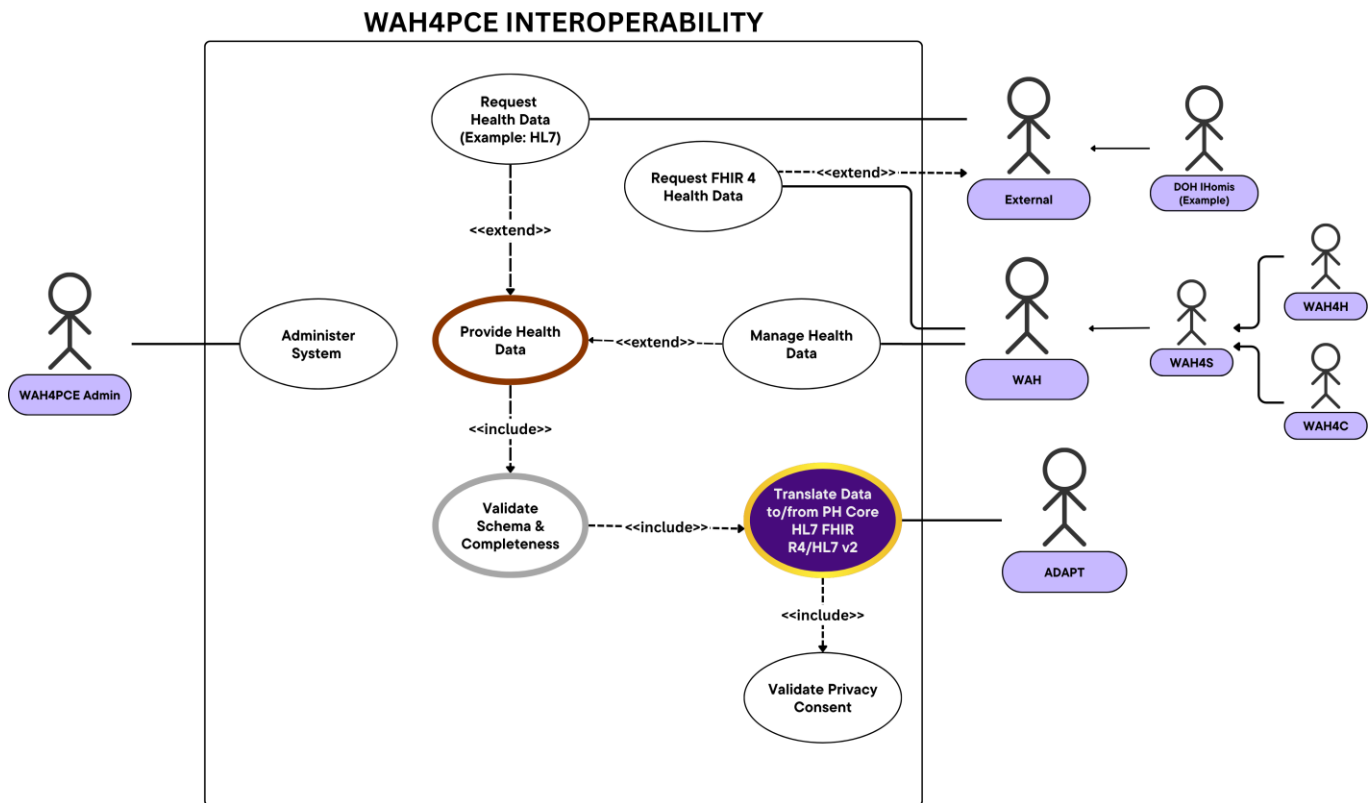


WAH4PCE INTEROPERABILITY

Use Case Full Description

Modular Use Case Diagram



UC-01 — Administer System

Scenario	The WAH4PCE Admin uses the IPaaS management console to configure API endpoints, manage tenant credentials, monitor system health, test data mappings, and review audit logs
Triggering Event	The WAH4PCE Admin opens the IPaaS administration portal and logs in.
Brief Description	This use case covers the main administration of the platform. The admin can manage security settings, routing rules, system settings, and system health. The interface features a continuous, non-linear tabbed navigation loop allowing the admin to freely switch between the Dashboard, Transaction Logs, and Data Mapper at any point without completing a strict sequence. Secure login is required.
Actors	WAH4PCE Admin
Related Use Cases	None
Stakeholders	WAH4PCE Admin: Needs secure access to system settings and monitoring information. All Tenants/Users: Depend on the admin to keep the platform available and health data secure.
Preconditions	The WAH4PCE Admin has valid administrative credentials in the identity provider.
Postconditions	The admin is authenticated. Valid configuration changes are applied to the gateway and recorded in the audit trail.

Flow of Activities

Actor	System
1. Opens the IPaaS login portal and enters the required administrative credentials, including MFA when required.	1.1 Checks the credentials with the secure identity provider and creates an access token. 1.2 Displays the administration dashboard.
2. Navigates to the Transaction Logs tab to review the complete audit trail of all data transformations.	2.1 Displays a table of transaction records showing the Transaction ID, Direction, Formats, Sent, Received, Status, Date.
3. Clicks the "All Statuses" dropdown to select a state filter (e.g., Success, Pending, Transforming, Quarantined) or the "All Sources" dropdown to select a system filter (e.g., iHOMIS (DOH), WAH Hospital).	3.1 Prepares the selected filter parameters for the log query.
4. Clicks the "Refresh" button.	4.1 Queries the system based on the selected filters and refreshes the displayed list of transaction records in the table.
5. Navigates to the Data Mapper in the dashboard to view a transaction (e.g., WAH to iHOMIS)	5.1 Displays a side-by-side view of the raw input and transformed output, with both payloads SHA-256 hashed by default.
6. Clicks the transaction dropdown to select a specific transaction log (e.g., 70ee4aa9 — iHOMIS → WAH (SUCCESS) or 81b27bea — WAH → iHOMIS (SUCCESS))	6.1 Identifies the selected transaction record and update the view.
7. Clicks the "Refresh" button in the Data Mapper	7.1 Fetches the latest transaction data and displays a side-by-side view of the raw input and transformed output, with both payloads SHA-256 hashed by default.
8. Clicks the "Unhash" button on the payloads	8.1 Reveals the JSON payloads side-by-side, displaying the raw input (e.g., WAH FHIR R4 BUNDLE) alongside the AI-transformed output (e.g., iHOMIS FORMAT) for visual verification
9. Clicks the "Comparison Table" button in the Data	9.1 Switches the display from the JSON view to a side-

Mapper view	by-side tabular comparison, detailing the specific fields, values, and mapping statuses (e.g., FILLED or EMPTY) for the sent and received data.
10. Navigates to Dashboard	10.1 Displays aggregate metrics and Recent Activity table
11. Clicks the "Logout" button to securely exit the administration portal.	11.1 Invalidates the active session token and redirects the admin back to the Login page.

Exception Conditions

- 1.1 Invalid Credentials:** If the credentials are incorrect or MFA fails, access is denied, the login attempt is blocked, the security event is logged, and the user is looped back to the Login page to try again.
- 1.2 Exceeded Login Attempts (Account Lockout):** If the user fails authentication 3 consecutive times, access is blocked, a security event is logged, the account/IP is temporarily locked to prevent brute-force attacks, and the login process terminates.

UC-02 — Provide Health Data

Scenario	A health-data transaction enters the IPaaS and uses the core pipeline for processing and routing.
Triggering Event	An actor starts a data transaction through one of the use cases that extends this core pipeline, such as Request Health Data, Request FHIR 4 Health Data, or Manage Health Data
Brief Description	This is the main health-data processing pipeline. It facilitates bidirectional data exchange by receiving the transaction, checking the data, and routing it to the correct destination (e.g., from WAH to iHOMIS, or iHOMIS to WAH). Other data operations use this pipeline.
Actors	System (triggered by External or WAH actors)
Related Use Cases	Request Health Data (Example: HL7) (<<extend>>) Request FHIR 4 Health Data (<<extend>>) Manage Health Data (<<extend>>) Validate Schema & Completeness (<<include>>)
Stakeholders	External / WAH Actors: Depend on the pipeline to process and route data reliably and securely. Patients: Depend on the system to handle their clinical data safely.
Preconditions	An authorized External or WAH tenant has started an API transaction.
Postconditions	The data is processed, translated when needed, and routed to the correct destination.

Flow of Activities

Actor	System
1. Starts a bidirectional transaction through an extending use case.	1.1 Starts the core processing pipeline and checks the incoming API payload. 1.2 Includes Validate Schema & Completeness to check the data. 1.3 Includes Validate Privacy Consent to verify consent signed flag. 1.4 Checks if source and destination formats differ. If transformation is required, includes Translate Data (UC-04) ; otherwise, bypasses transformation and advances the raw validated payload directly to status dispatch 1.5 Updates transaction status to SUCCESS, dispatches the payload via webhook, and returns HTTP 200 OK.

Exception Conditions

1.1 Pipeline Failure: If the core processing engine has a critical failure, the transaction stops, HTTP 500 Internal Server Error is returned, and the WAH4PCE Admin is alerted through system telemetry.

UC-03 — Validate Schema & Completeness

Scenario	The system checks the structure of the incoming JSON/XML data and verifies that required fields are present before the data is processed further.
Triggering Event	The system automatically starts this use case when Provide Health Data receives a payload.
Brief Description	This internal validation step checks the payload against the registered schemas and required fields. If the data is valid, processing can continue to translation.
Actors	System (Internal)
Related Use Cases	Provide Health Data (Included by) Translate Data to/from PH Core HL7 FHIR R4/HL7 v2 (<<include>>)
Stakeholders	All Tenants: Depend on validation to prevent malformed or incomplete data from moving to downstream systems.
Preconditions	A data payload is active in the core processing pipeline.
Postconditions	The payload is confirmed to be structurally valid and processing can continue.

Flow of Activities

Actor	System
None (System-driven)	1.1 Receives the payload from the core pipeline. 1.2 Validates required routing fields (source_system, source_format, destination_system, destination_format). 1.3 Sends a validation-success confirmation back to the core pipeline.

Exception Conditions

1.2 Validation Failure: If the payload is malformed or fails the required checks, the transfer is stopped, the event is logged, and HTTP 400 Bad Request is returned to the caller.

UC-04 — Translate Data to/from PH Core HL7 FHIR R4/HL7 v2

Scenario	Health data is transformed and mapped so that it follows the PH Core HL7 FHIR R4 or legacy HL7 v2 format.
Triggering Event	The system starts this use case when the core pipeline determines that data transformation is required and privacy consent has been verified.
Brief Description	This use case uses ADAPT to transform and map health-data structures between supported formats. The goal is to keep the data understandable and interoperable between different systems while applying the required privacy checks.
Actors	ADAPT
Related Use Cases	Provide Health Data (Included by) Validate Privacy Consent (<<include>>)
Stakeholders	ADAPT System: Needs the correct mapping rules to transform the data. Data Consumers: Need standardized data that can be used by their systems.
Preconditions	A validated payload is ready for transformation.
Postconditions	The payload is transformed into the required standard without changing its clinical meaning.

Flow of Activities

Actor	System
None (System-driven)	1.1 Sends the validated payload to the ADAPT AI transformation engine. 1.2 ADAPT analyzes the source format and attempts to apply the required AI mapping rules. 1.3 If the AI transformation fails or encounters an issue, the system triggers a fallback and switches to a deterministic algorithmic transformation to map the data. 1.4 Completes the transformation and returns the standardized payload to the pipeline.

Exception Conditions

1.2 Mapping Failure: If ADAPT cannot safely transform the data without changing its clinical meaning, the Payload is marked as QUARANTINED, transaction stops, and returns HTTP 422.

UC-05 — Validate Privacy Consent

Scenario	Before transformed health data is released, the system checks the patient's current consent to make sure the data-sharing action is allowed.
Triggering Event	The system automatically starts this use case during Translate Data to/from PH Core HL7 FHIR R4/HL7 v2.
Brief Description	This is the privacy check before data can continue. The system uses the patient identifier and consent information to decide whether the transaction is allowed.
Actors	System (Internal)
Related Use Cases	Translate Data to/from PH Core HL7 FHIR R4/HL7 v2 (Included by)
Stakeholders	Patients: Need their privacy choices and opt-out instructions to be followed. Compliance Officers: Need a reliable and complete record of consent decisions.
Preconditions	The patient identifier can be read, and the consent registry is available.
Postconditions	The consent result is added to the transaction, and the decision is recorded.

Flow of Activities

Actor	System
None (System-driven)	1.1 Reads the consent_signed flag from the transaction payload. 1.2 Checks the consent registry for the applicable sharing rules. 1.3 Compares the consent rules with the current transaction. 1.4 Sends an approval token to ADAPT when the transaction is allowed.

Exception Conditions

1.3 Consent Denied: If the patient has opted out or the applicable policy does not allow the transaction, the Payload is marked as QUARANTINED, transaction stops, and returns HTTP 422.

UC-06 — Request Health Data (Example: HL7)

Scenario	An external tenant requests specific patient records from the IPaaS in legacy HL7 format.
Triggering Event	A requesting actor sends a health-data query to the integration endpoint
Brief Description	This bidirectional use case allows an authorized system to retrieve health data using the legacy HL7 format, pending approval from the source organization. Either an external tenant (like iHOMIS) can request data from WAH, or WAH can request data from an external tenant.
Actors	External (e.g., DOH iHomis)
Related Use Cases	Provide Health Data (<<extend>>)
Stakeholders	External Actors: Need accurate patient data for reporting or continuity of care.
Preconditions	The External actor is authenticated and authorized for the requested data scope, using the supported authentication method such as OAuth2 or API Key.
Postconditions	The requested HL7 data is returned and the retrieval is recorded in the audit trail.

Flow of Activities

Actor	System
1. IHOMIS Sends a health-data query through the API gateway..	1.1 Authenticates the actor's identity and forwards the request to the target organization's queue for review.
2. (Source Authorized User) Reviews the incoming bidirectional request in their respective portal (e.g., WAH or iHOMIS) and clicks approve.	2.1 Extracts the requested patient records from the local source database. 2.2 Uses the Provide Health Data pipeline to transform the records and route the formatted HL7 response back to the requesting actor.
3. (Requesting Actor) Receives the formatted legacy HL7 response and ingests it into their local system.	3.1 Closes the transaction session and records successful receipt in the audit trail.

Exception Conditions

1.1 Request Declined: If the WAH authorized user rejects the request, the transaction stops, and a rejection notification is sent to the requesting actor.

2.1 Data Not Found: If no matching record exists in the WAH database upon approval, the system returns HTTP 404 Not Found.

UC-07 — Request FHIR 4 Health Data

Scenario	A tenant of WAH requests structured FHIR R4 clinical resources from the IPaaS.
Triggering Event	An External actor sends a RESTful GET or POST request for FHIR R4 resources..
Brief Description	This bidirectional use case allows an authorized system to securely retrieve modern, structured health data using FHIR R4, pending approval from the source organization. Either an external tenant (like iHOMIS) can request data from WAH, or WAH can request data from an external tenant.
Actors	External, WAH (e.g., WAH4H, WAH4C)
Related Use Cases	Provide Health Data (<<extend>>)
Stakeholders	WAH / External Actors: Need structured, REST-based health data for modern clinical applications.
Preconditions	The requesting actor is authenticated and authorized for the requested data scope, using the supported authentication method such as a valid access token.
Postconditions	The requested FHIR R4 JSON bundle is securely returned via webhook to the requesting system, and the retrieval is recorded in the audit trail.

Flow of Activities

Actor	System
1. (Requesting Actor) Sends a REST request for the required FHIR R4 resources through the API gateway.	1.1 Authenticates the actor's identity and forwards the request to the target organization's queue for review.
2. (Source Authorized User) Reviews the incoming bidirectional request in their respective portal (e.g., WAH or iHOMIS) and clicks approve.	2.1 Extracts the requested patient records from the local source FHIR repository. 2.2 Uses the Provide Health Data pipeline to protect the data in transit using TLS and route the structured FHIR response back to the requesting actor.
3. (Requesting Actor) Receives the formatted FHIR R4 JSON bundle via webhook / client endpoint and updates the patient's local clinical record.	3.1 Closes the transaction session and records successful receipt in the audit trail.

Exception Conditions

1.1 Request Declined: If the source authorized user rejects the request, the transaction stops, and a rejection notification is sent to the requesting actor.

2.1 Data Not Found: If no matching record exists in the source FHIR repository upon approval, the system returns HTTP 404 Not Found.

UC-08 — Manage Health Data

Scenario	An authorized WAH actor updates, archives, or amends existing clinical records through the IPaaS.
Triggering Event	A WAH actor sends an HTTP PUT, POST, or PATCH request to modify an existing record.
Brief Description	This use case allows authorized WAH actors to manage their submitted health data while keeping the data consistent and maintaining version history.
Actors	WAH (e.g., WAH4H, WAH4S, WAH4C)
Related Use Cases	Provide Health Data (<<extend>>)
Stakeholders	WAH Actors: Need to update patient information securely when clinical information changes.
Preconditions	The target record exists, and the WAH actor has permission to write to that record.
Postconditions	The record is updated, its version history is maintained, and the change is recorded.

Flow of Activities

Actor	System
1. Sends an update request through the API gateway.	1.1 Checks the actor's write permission for the requested resource. 1.2 Uses the Provide Health Data pipeline to process the update. 1.3 Applies the update, creates the new record version, and returns HTTP 200 OK or 204 No Content.

Exception Conditions

1.3 Record Locked / Conflict: If the record is locked or the supplied version does not match the current version, such as an eTag mismatch, the system rejects the update with HTTP 409 Conflict.